

Degree, Semester & Branch: V Semester B.E. Computer Science and Engineering.

Course Code & Title: CS8592 Object Oriented Analysis & Design

Name of the Faculty member: Mr. D.Lakshmanan, Assistant Professor (S.G)/CSE

Innovative practice: Think-pair-share

Topic: Use Case diagram for real time systems.

Objectives:

- Students should be able to demonstrate conceptual understanding.
- Students should be able to analyse the pros and cons of different ways of coding.

Justification for choosing this topic:

Collaborative, active learning strategy, in which students work on a problem posed by instructor, first individually (Think), then in pairs (Pair) or groups, and finally together with the entire class (Share).

- T (Think): Teacher asks a specific question about the topic. Students "think" about what they know or have learned, and come up with their own individual answer to the question. [Takes 1-3 Minutes].
- P (Pair): Teacher asks another question, related to the previous one that is suitable to deepen the students' understanding of the topic. Each student is paired with another student. They share their thinking with each other and proceed with the task. [Takes 5-10 Minutes].
- S (Share): Students share their thinking (or solution) with the entire class. Teacher moderates the discussion and highlights important points. [Takes 10-20 minutes].

Benefit of TPS

- Students are actively engaged.
- Students learn from each other.
- Makes class interactive.
- Builds a friendly, yet academic atmosphere.
- Includes all the students in the teaching-learning process.

Activity:

During the last twenty minutes of fifth hour on 04.07.2019 Think-Pair-Share innovative practice conducted for III year CSE students, after explained the concept of use case and its elements with an example of ATM system.

First, I asked the students to think about the real time system such as Monopoly Game System, Next Gen POS System, Hostel Leave Management System, College Inward-Outward

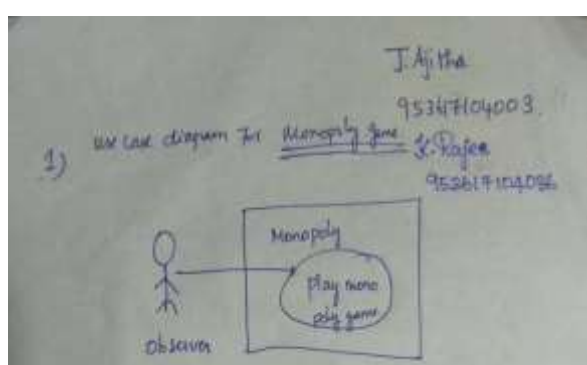
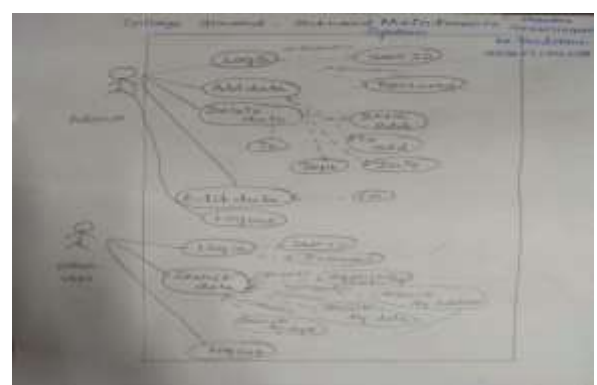
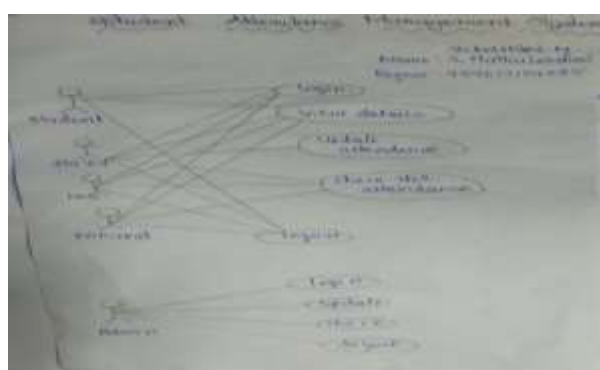
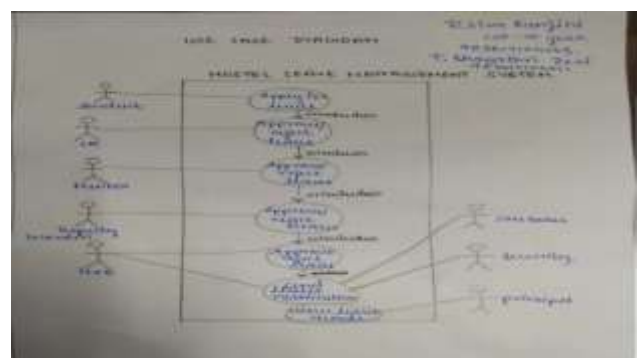
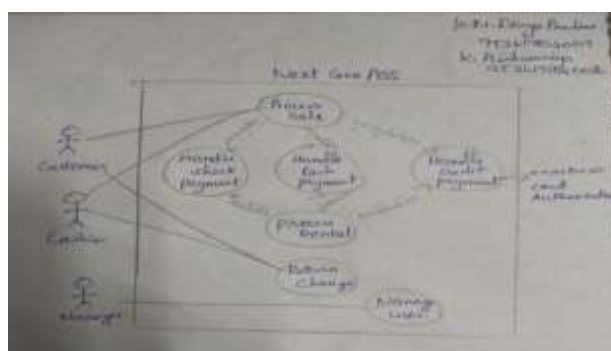
Maintenance System and Student Attendance Management System for drawing the use case diagram.

Then I make them as a pair based on the topics they chosen to share their thinking with partner.

Finally I asked the one of the pair to explain the concept to whole class for further discussion. Ms. J.Ajitha, III year CSE students share her knowledge of Monopoly Game Use case diagram to all students.



During the session Ms. J.Ajitha share the knowledge of Monopoly Game Use Case Diagram



This activity helps in attaining the following CO – PO mapping:

Course Outcome / Programme Outcome	PO1	PO2	PO3	PO6
State the usage of different UML diagrams and Unified process.	2	2	3	1

Reflection critique

Challenges

- Some of the students were not able to easily draw, even the basic use case diagram.
- Students felt very difficult in this activity and could not complete in time whenever new real time system asked in the question.
- Students are hesitating to speak in front of all.

Steps taken for address the challenges

- I motivated the students to involve in activity effectively by means of combine one bright student and one slow learner as a pair, so that slow learner can able to clarify the doubt, they can better understand about the topic. After that I combine two average students as a pair, so that they can share their knowledge with them and come up with better understanding about the topic.
- Different types of real time system are discussed in the class room, so students come to know how to draw the use case diagram whenever new real time system is asked in the question.
- I motivated the students to speak in front of all the students and share their knowledge to others student.

SESSION OUTCOMES:

- This makes the students to draw the usage of use case diagram for real time system.
- It provides opportunities for the students to acquire knowledge about various real time systems, its Actors, use cases and its association.

References:

<http://www.et.iitb.ac.in/TeachingResources.html>

Prepared by:

Mr. D.Lakshmanan, AP(S.G)/CSE

Approved by:

Dr. K.Vijayalakshmi, HOD/CSE